

JONATHAN ONG CHONG AN

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Availability: January 2027 – August 2027
Singapore, SG

EDUCATION

Bachelor of Science, Information Systems (Product Development) Jan 2023 – Dec 2027
Singapore Management University | Second Major: Computer Science (Artificial Intelligence)

SUMMARY

Information Systems undergraduate with experience in enterprise GenAI, RAG, and production data systems development. Skilled in building reliable AI workflows with strong emphasis on grounding, testing, data quality, and responsible deployment.

EXPERIENCE

AI & Data Engineering Intern, Inland Revenue Authority of Singapore (IRAS) May 2026 – Nov 2026

- Migrated 10+ legacy SAS ETL scripts to PySpark for monthly batch-processing workflows handling approximately 5–10 million tax records, supporting downstream analytics and reporting.
- Developed reusable validation checks across schemas, joins, timestamps, and business rules to identify data discrepancies and improve confidence in migrated pipeline outputs.
- Integrated internal search APIs into an enterprise assistant’s retrieval workflow, improving access to trusted organisational knowledge and supporting more relevant, evidence-grounded responses.
- Supported the migration of internal machine-learning APIs to Azure ML by standardising model artefact management, strengthening logging and exception handling, and introducing automated pytest checks that increased the CI test success rate by approximately 50%.

DevOps Engineer Intern, YTL PowerSeraya Jan 2026 – Apr 2026

- Built and rolled out Azure DevOps multi-stage YAML pipelines for 4 internal gas-management applications, reducing deployment time from 60 to 20 minutes and improving release consistency.
- Designed a self-hosted agent approach for secure IaaS deployments within an isolated VNet, enabling CI/CD execution on application VMs and documenting the workflow for developer adoption.

Cloud Engineer Intern, Global Enterprise Mobility (GEM) May 2025 – Aug 2025

- Built Jenkins CI/CD pipelines using Terraform-provisioned AWS infrastructure and GCP Secret Manager, reducing release time from 40 to 20 minutes while improving deployment repeatability and credential security.
- Duplicated webhook events using a DynamoDB idempotency layer, resolving a production defect that affects approximately 12–15% of transactions.

PROJECTS

PitchPerfect — AI Evaluation Platform | Product Owner & Data Pipeline Lead Jun 2026 – Present

- Product Owner in a six-member Agile Scrum team developing an AI evaluation platform for the Lee Kuan Yew Global Business Plan Competition, reducing the time it takes to evaluate pitch decks by 80%.
- Own the ingestion and retrieval workstream, translating a manual pitch-deck review process into structured feature extraction, evidence retrieval, and readiness-assessment stages.
- Design a pipeline to ingest labeled pitch decks, extract rubric-aligned features and metadata, generate embeddings, and retrieve supporting evidence for consistent evaluation.

YilongMa — Retrieval-Augmented Generation Data Pipeline Aug 2025 – Dec 2025

- Built an end-to-end data and retrieval pipeline using Python, ChromaDB, Llama 3.2 1B, and hybrid search to generate grounded responses from a transcript-based knowledge base.
- Processed more than 50 hours of audio using Whisper and developed evaluation notebooks to assess retrieval relevance and response grounding.

Wander — Full-Stack Travel Planning Application Sep 2024 – Nov 2024

- Built a full-stack travel planning application using Vue.js and Flask, integrating REST APIs and the OpenAI API to generate personalised itineraries based on user preferences.
- Implemented dynamic location suggestions from submitted TikTok URLs, contributing to an A+ project grade.

CERTIFICATIONS

AWS Certified Cloud Practitioner | Google IT Automation with Python | Java Certified Foundations Associate

SKILLS

Languages: Python, SQL, Java, JavaScript, Bash
Software Engineering: FastAPI, Flask, Vue.js, React.js, REST APIs, Pytest, Git
Data Engineering: PySpark, Pandas, Databricks, Azure Synapse, ETL, Data Validation
Cloud & Infrastructure: AWS, Azure, Google Cloud Platform, Docker, Terraform, Jenkins, CICD
AI/ML: RAG, Embeddings, Hybrid Retrieval, LLM Evaluation, Prompt Engineering, Agent Harness